

# Debojyoti Ghosh — Publications

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## JOURNAL ARTICLES (29)

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- de Jong, E. K., Gunawardena, N., Katona, J. E., Beydoun, H., **Ghosh, D.**, Caldwell, P., *Data-Driven Reduced Order Modeling for Warm Rain Microphysics*, Journal of Geophysical Research: Machine Learning and Computation, 3 (3), 2026, e2025JH001103, doi:10.1029/2025JH001103.
- Tsai, P.-H., Chung, S. W., **Ghosh, D.**, Loffeld, J., Choi, Y., Belof, J. L., *Local reduced-order modeling for electrostatic plasmas by physics-informed solution manifold decomposition*, Computer Physics Communications, 322, 2026, 110039, doi:10.1016/j.cpc.2026.110039.
- Michel, P., Oudin, A., Rajesh, H., Ou, K., Chakraborty, D., Cao, S., Kur, E., Lancia, L., **Ghosh, D.**, Riconda, C., Wurtele, J. S., Edwards, M. R., *Entropy-mode-driven gas optics*, Physical Review Applied, 25, 2026, 024037, doi:10.1103/pnhn-w6tl.
- Salian, L. V., Samala, R., **Ghosh, D.**, *Central compact finite-difference scheme with high spectral resolution for KdV equation*, Numerical Methods for Partial Differential Equations, 42 (1), 2026, e70060, doi:10.1002/num.70060.
- Dorf, M., Dorr, M., Geyko, V., **Ghosh, D.**, Umansky, M., Angus, J., *Semi-implicit continuum kinetic modeling of weakly collisional parallel transport in a magnetic mirror*, Physics of Plasmas, 32, 2025, 103903, doi:10.1063/5.0293154.
- van de Wetering, J., Angus, J. R., Farmer, W., Geyko, V., **Ghosh, D.**, Grote, D., Weber, C., Zimmerman, G., *Particle-in-cell simulations of burning inertial confinement fusion capsule implosions*, Physical Review E, 112, 2025, 045207, doi:10.1103/zwjx-jbxl.
- Dorf, M., Dorr, M., **Ghosh, D.**, Umansky, M., Soukhanovskii, V., *Implicit full-F simulations of neoclassical ion transport*, Physics of Plasmas, 32, 2025, 083904, doi:10.1063/5.0280028.
- Oudin, A., **Ghosh, D.**, Riconda, C., Lancia, L., Kur, E., Ou, K., Perez-Ramirez, V. M., Lee, J., Edwards, M. R., Michel, P., *PIAFS: A 2D nonlinear hydrodynamics code to model gaseous optics*, Physics of Plasmas, 32, 2025, 072714, doi:10.1063/5.0268318.
- Geyko, V. I., Joseph, I., Dorf, M. A., **Ghosh, D.**, Dorr, M. R., *Drift kinetic electrostatic simulations of the edge localized mode heat pulse*, Physics of Plasmas, 31, 2024, 123903, doi:10.1063/5.0230913.
- Angus, J. R., Farmer, W., Friedman, A., Geyko, V., **Ghosh, D.**, Grote, D., Larson, D., Link, A., *An implicit particle code with exact energy and charge conservation for studies of dense plasmas in axisymmetric geometries*, Journal of Computational Physics, 519, 2024, 113427, doi:10.1016/j.jcp.2024.113427.
- Dorf, M., Dorr, M., **Ghosh, D.**, *Development of an implicit electromagnetic capability for a hybrid gyrokinetic ion-fluid electron model*, Contributions to Plasma Physics, 64 (7-8), 2024, doi:10.1002/ctpp.202300114.
- Bonneville, C., Choi, Y., **Ghosh, D.**, Belof, J. L., *GPLaSDI: Gaussian Process-based interpretable Latent Space Dynamics Identification through deep autoencoder*, Computer Methods in Applied Mechanics and Engineering, 418 (A), 2024, 116535, doi:10.1016/j.cma.2023.116535.
- Angus, J. R., Farmer, W., Friedman, A., **Ghosh, D.**, Grote, D., Larson, D., Link, A., *An implicit particle code with exact energy and charge conservation for electromagnetic studies of dense plasmas*, Journal of Computational Physics, 491, 2023, 112383, doi:10.1016/j.jcp.2023.112383.
- Kim, Y., **Ghosh, D.**, Constantinescu, E. M., Balakrishnan, R., *GPU-accelerated DNS of compressible turbulent flows*, Computers and Fluids, 251, 2023, 105744, doi:10.1016/j.compfluid.2022.105744.
- Angus, J. R., Link, A., Friedman, A., **Ghosh, D.**, Johnson, J. D., *On numerical energy conservation for an implicit particle-in-cell method coupled with a binary Monte-Carlo algorithm for Coulomb collisions*, Journal of Computational Physics, 456, 2022, doi:10.1016/j.jcp.2022.111030.
- Chapman T., Winjum B. J., Berger, R. L., Dimits, A., Banks, J. W., Brunner, S., Joseph, I., **Ghosh, D.**, *Nonlinear kinetic simulation study of the ion-ion streaming instability in single- and multi-ion species plasmas*, Physics of Plasmas, 28, 022105, 2021, doi:10.1063/5.0015302.
- Ghosh, D.**, Chapman, T. D., Berger, R. L., Dimits, A., Banks, J. W., *A Multispecies, Multifluid Model for Laser-Induced Counterstreaming Plasma Simulations*, Computers and Fluids, 186, 2019, 38-57, doi:10.1016/j.compfluid.2019.04.012.
- Dimits, A. M., Banks, J. W., Berger, R. L., Brunner, S., Chapman, T. D., Copeland, D., **Ghosh, D.**, Arrighi, W. J., Hittinger, J., Joseph, I., *Linearized Coulomb Collision Operator for Simulation of Interpenetrating Plasma Streams*, IEEE Transactions on Plasma Science, 47 (5), 2019, 2074-2080, doi:10.1109/TPS.2019.2897790.
- Ghosh, D.**, Dorf, M. A., Dorr, M. R., Hittinger, J., *Kinetic Simulation of Collisional Magnetized Plasmas with Semi-Implicit Time Integration*, Journal of Scientific Computing, 77 (2), 2018, 819-849, doi:10.1007/s10915-018-0726-6.
- Dorr, M., Colella, P., Dorf, M., **Ghosh, D.**, Hittinger, J., Schwartz, P. O., *High-order Discretization of a Gyrokinetic Vlasov Model in Edge Plasma Geometry*, Journal of Computational Physics, 373, 2018, 605-630, doi:10.1016/j.jcp.2018.07.008.

- Dorf, M., Dorr, M., Hittinger, J., Lee, W., **Ghosh, D.**, *High-order finite-volume modeling of drift waves*, Journal of Computational Physics, 373, 2018, 446-454, doi:10.1016/j.jcp.2018.07.009.
- Ghosh, D.**, Constantinescu, E. M., *Well-Balanced, Conservative Finite-Difference Algorithm for Atmospheric Flows*, AIAA Journal, 54 (4), 2016, 1370-1385, doi:10.2514/1.J054580.
- Ghosh, D.**, Constantinescu, E. M., *Semi-Implicit Time Integration of Atmospheric Flows with Characteristic-Based Flux Partitioning*, SIAM Journal on Scientific Computing, 38 (3), 2016, A1848-A1875, doi:10.1137/15M1044369.
- Wang, P., Barajas-Solano, D. A., Constantinescu, E. M., Abhyankar, S., **Ghosh, D.**, Smith, B. F., Huang, Z., Tartakovsky, A. M., *Probabilistic Density Function Method for Stochastic ODEs of Power Systems with Uncertain Power Input*, SIAM/ASA Journal on Uncertainty Quantification, 3 (1), 2015, 873-896, doi:10.1137/130940050.
- Ghosh, D.**, Constantinescu, E. M., Brown, J., *Efficient Implementation of Nonlinear Compact Schemes on Massively Parallel Platforms*, SIAM Journal on Scientific Computing, 37 (3), 2015, C354-C383, doi:10.1137/140989261.
- Ghosh, D.**, Baeder, J. D., *Weighted Non-Linear Compact Schemes for the Direct Numerical Simulation of Compressible, Turbulent Flows*, Journal of Scientific Computing, 61 (1), 2014, 61-89, doi:10.1007/s10915-014-9818-0.
- Ghosh, D.**, Medida, S., Baeder, J. D., *Application of Compact-Reconstruction Weighted Essentially Nonoscillatory Schemes to Compressible Aerodynamic Flows*, AIAA Journal, 52 (9), 2014, 1858-1870, doi:10.2514/1.J052654.
- Ghosh, D.**, Baeder, J. D., *A High-Order Accurate Incompressible Navier Stokes Algorithm for Vortex Ring Interactions with Solid Wall*, AIAA Journal, 50 (11), 2012, 2408-2422, doi:10.2514/1.J051537.
- Ghosh, D.**, Baeder, J. D., *Compact Reconstruction Schemes with Weighted ENO Limiting for Hyperbolic Conservation Laws*, SIAM Journal on Scientific Computing, 34 (3), 2012, A1678-A1706, doi:10.1137/110857659.

## BOOK CHAPTERS (1)

- Ghosh, D.**, Constantinescu, E. M., *Nonlinear Compact Finite-Difference Schemes with Semi-Implicit Time Stepping*, in Spectral and High Order Methods for Partial Differential Equations ICOSAHOM 2014, Springer Lecture Notes in Computational Science and Engineering, Volume 106, 2015, 237-245, doi:10.1007/978-3-319-19800-2\_20.

## PEER - REVIEWED CONFERENCE PAPERS (12)

- Tsai, P.-H., Chung, S. W., **Ghosh, D.**, Loffeld, J., Choi, Y., Belof, J. L., *Accelerating Kinetic Simulations of Electrostatic Plasmas with Reduced-Order Modeling*, Workshop on Machine Learning and the Physical Sciences, 37th conference on Neural Information Processing Systems (NeurIPS), December 10 - 16, 2023, New Orleans, LA.
- Bonneville, C., Choi, Y., **Ghosh, D.**, Belof, J. L., *Data-Driven Autoencoder Numerical Solver with Uncertainty Quantification for Fast Physical Simulations*, Workshop on Machine Learning and the Physical Sciences, 37th conference on Neural Information Processing Systems (NeurIPS), December 10 - 16, 2023, New Orleans, LA.
- Dorf, M., Dorr, M., **Ghosh, D.**, Umansky, M., *A hybrid gyrokinetic ion – fluid electron model for edge plasma simulations*, 29th IAEA Fusion Energy Conference, October 16 - 21, 2023, London, UK.
- Christopher, J., Davis, A., Gao, X., **Ghosh, D.**, Dorf, M., Dorr, M., Ricketson, L., *On Preconditioning for the Two-Field Neutral Model*, AIAA Scitech 2022 Forum, January 3 - 7, 2022, San Diego, CA, doi:10.2514/6.2022-1108.
- Munafo, A., Vogiatzis, K., **Ghosh, D.**, Vedula, P., Panesi, M., Josyula, E., *High-Order Techniques for Multi-Component Turbulent Non-Equilibrium Hypersonic Flows*, AIAA Scitech 2020 Forum, January 6 - 10, 2020, Orlando, FL, doi:10.2514/6.2020-2193.
- Dorf, M. A., Dorr, M., **Ghosh, D.**, Hittinger, J., Umansky, M. V., Angus, J., Pankin, A., *Simulation of Cross-Separatrix Edge Plasma Transport with the Continuum Gyrokinetic Code COGENT*, 27th IAEA Fusion Energy Conference, October 22 - 27, 2019, Gandhinagar, Gujarat, India.
- Guhur, P.-L., Constantinescu, E. M., **Ghosh, D.**, Peterka, T., Cappello, F., *Detection of Silent Data Corruption in Adaptive Numerical Integration Solvers*, IEEE Cluster Conference, September 5-8, 2017, Honolulu, HI, doi:10.1109/CLUSTER.2017.13.
- Ghosh, D.**, Dorf, M., Hittinger, J., Dorr, M., *Implicit-Explicit Time Integration for the Vlasov-Fokker-Planck Equations*, AIAA Paper 2017-4016, 48th AIAA Plasmadynamics and Lasers Conference, June 5-9, 2017, Denver, CO, doi:10.2514/6.2017-4016.
- Ghosh, D.**, Constantinescu, E. M., *Well-Balanced Formulation of Gravitational Source Terms for Conservative Finite-Difference Atmospheric Flow Solvers*, AIAA Paper 2015-2889, 7th AIAA Atmospheric and Space Environments Conference, June 22-26, 2015, Dallas, TX, doi:10.2514/6.2015-2889.
- Ghosh, D.**, Medida, S., Baeder, J. D., *Compact-Reconstruction Weighted Essentially Non-Oscillatory Schemes for Unsteady Navier-Stokes Equations*, AIAA Paper 2012-2832, 42nd AIAA Fluid Dynamics Conference and Exhibit, June 25-28, 2012, New Orleans, LA, doi:10.2514/6.2012-2832.
- Ghosh, D.**, Baeder, J. D., *Numerical Simulation of Vortex Ring Interactions with Solid Wall*, 49th AIAA Aerospace Sciences Meeting, Jan 4-7,

2011, Orlando, FL, doi:10.2514/6.2011-675.

**Ghosh, D.**, Baeder, J. D., *A High Order Conservative Upwind Algorithm for the Incompressible Navier Stokes Equations*, AIAA Paper 2010-5030, 40th AIAA Fluid Dynamics Conference and Exhibit, June 28 – July 1, 2010, Chicago, IL, doi:10.2514/6.2010-5030.

## TECHNICAL REPORTS (2)

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Terasaki, G., **Ghosh, D.**, Navid, A., *Improved School and Workplace Interactions in ExaEpi, an Agent-Based Epidemiological Model*, Lawrence Livermore National Laboratory, Livermore, CA, 2025.

**Ghosh, D.**, Constantinescu, E. M., Brown, J., *Scalable Nonlinear Compact Schemes*, Technical Memorandum, ANL/MCS-TM-340, Argonne National Laboratory, April 2014.

## THESES (3)

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*Compact-Reconstruction Weighted Essentially Non-Oscillatory Schemes for Hyperbolic Conservation Laws*, Ph. D. Thesis, University of Maryland, College Park, January 2013.

*Higher Order Non-Oscillatory Schemes in Ideal Magnetohydrodynamics*, Masters Thesis, Indian Institute of Technology Bombay, July 2006.

*Finite Volume Time Domain Solutions of Maxwell's Equations*, Bachelors Seminar, Indian Institute of Technology Bombay, March 2004.

## INVITED AND MINISYMPOSIUM TALKS (35)

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**Ghosh, D.**, Beydoun, H., McGuffin, D., Gardner, D. J., Lundquist, K., *A scalable particle-based microphysics model for atmospheric flow simulations*, 18th U.S. National Congress on Computational Mechanics (USNCCM), Minisymposium on Computational fluid dynamics (CFD) and fluid-structure interaction (FSI): method development and applications, July 20 - 24, 2025, Chicago, IL.

Angus, J. R., Farmer, W., Friedman, A., Fu, Y., Geyko, V., **Ghosh, D.**, Grote, D., Larson, D., Ludwig, J., Van de Wetering, J., Zimmerman, G., Link, A., *An implicit particle code for fully kinetic simulations of dynamically compressing plasmas*, IEEE Pulsed Power and Plasma Science Conference (PPPS 2025), Session on Computational Plasmas, June 15 - 20, 2025, Berlin, Germany.

**Ghosh, D.**, Beydoun, H., McGuffin, D., Gardner, D. J., Lundquist, K., *Super Droplet Method for Lagrangian Microphysics Simulations in ERF*, SIAM Conference on Computational Science and Engineering, Minisymposium on Energy Research and Forecasting (ERF): A New Performance-Portable Atmospheric Modeling Code, March 3 - 7, 2025, Fort Worth, TX.

Myers, A., Nugent, P., Singhal, A., Acquesta, E., **Ghosh, D.**, Keilbart, N., Nguyen, T., *ExaEpi: An Exascale-Capable Agent-Based Model for Epidemiology*, SIAM Conference on Computational Science and Engineering, Minisymposium on Predictive Disease Modeling and Simulations for Decision-making, March 3 - 7, 2025, Fort Worth, TX.

Tsai, P.-H., Chung, S. W., **Ghosh, D.**, Loffeld, J., Choi, Y., Belof, J., *Accelerating Kinetic Simulations of Electrostatic Plasmas with Local Reduced-Order Modeling*, SIAM Conference on Computational Science and Engineering, Minisymposium on Scientific Machine Learning at Scale, March 3 - 7, 2025, Fort Worth, TX.

Dorf, M., Dorr, M., **Ghosh, D.**, *A Semi-Implicit Gyrokinetic Ion – Fluid Electron Hybrid Model for Edge Plasma Simulations*, Platform for Advanced Scientific Computing (PASC), Minisymposium on High Performance Computing for Magnetic Fusion Applications - Part II, June 3 - 5, 2024, Zurich, Switzerland.

**Ghosh, D.**, Dorf, M., Dorr, M., *Magnetized Plasma Simulations with High-Order Implicit-Explicit Time Integrators*, 9th European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS), Minisymposium on Computational Kinetic Theory II, June 3 - 7, 2024, Lisbon, Portugal.

Tsai, P.-H., Chung, S. W., **Ghosh, D.**, Loffeld, J., Choi, Y., Belof, J., *Projection-Based Reduced Order Model for Accelerating Kinetic Simulations of Electrostatic Plasmas*, SIAM Conference on Parallel Processing for Scientific Computing (PP), Minisymposium on HPC Algorithms for Kinetic Equations for Plasmas, March 5 - 8, 2024, Baltimore, MD.

Choi, Y., Copeland, D., Cheung, S. W., Kim, Y., He, X., Fries, W., Bonneville, C., **Ghosh, D.**, Shin, Y., Park, J. S., Belof, J., Afeyan, B., *Physics-Guided Latent Space Dynamics Identification in Model Order Reduction*, 17th U.S. National Congress on Computational Mechanics (USNCCM), Minisymposium on Model Order Reduction for Parameterized Continuum Mechanics, July 23 - 27, 2023, Albuquerque, NM.

**Ghosh, D.**, Dorf, M., Dorr, M., Ricketson, L., *A Multiphysics Implicit-Explicit Time Integration Algorithm for Simulating Tokamak-Edge Plasma Dynamics*, 17th U.S. National Congress on Computational Mechanics (USNCCM), Minisymposium on Advances in Numerical Methods for Solution of PDEs, July 23 - 27, 2023, Albuquerque, NM.

Bonneville, C., He, X., Chen, J.-S., Fries, W., Cheung, S. W., **Ghosh, D.**, Belof, J., Springer, K., Choi, Y., *Certified and Parameterized Latent Space Dynamics Identification for Time Dependent Image Data*, AAAI Spring Symposium on Computational Approaches to Scientific Discovery, March 27 - 29, 2023, San Francisco, CA.

**Ghosh, D.**, Dorr, M., Dorf, M., Ricketson, L., *Semi-Implicit Time Integration for Multiscale Tokamak-Edge Plasma Dynamics*, SIAM Conference

- on Computational Science and Engineering, Minisymposium on Computational Methods for Kinetic Models, February 26 - March 3, 2023, Amsterdam, The Netherlands.
- Bonneville, C., Choi, Y., **Ghosh, D.**, *GPLaSDI: Identifying Autoencoders Latent Space Dynamics with Gaussian Processes for Non-Intrusive Reduced Order Models*, SIAM Conference on Computational Science and Engineering, Minisymposium on Reduced Order Modeling of Differential Equations Through Deep Learning Algorithms, February 26 - March 3, 2023, Amsterdam, The Netherlands.
- High-Order Semi-Implicit Time Integration Methods for Multiscale Advective Physics*, University of Zaragoza, October 2022 (Online talk).
- Ricketson, L., Dorr, M., Dorf M., **Ghosh, D.**, *Continuum Kinetic Simulations of Tokamak Edge Plasmas with COGENT*, Platform for Advanced Scientific Computing (PASC) Conference, Minisymposium on High Performance Computing in Kinetic Simulations of Plasmas - Part III: Advanced Numerical Methods and Algorithms, June 27 - 29, 2022, Basel, Switzerland.
- Ricketson, L., **Ghosh, D.**, Dorr, M., Dorf M., *Block Preconditioning of a Semi-Implicit Gyrokinetic Model of Fusion Plasmas*, 17th Copper Mountain Conference On Iterative Methods, Session on Scalable Solvers for Coupled Multiphysics 2, April 4 - 8, 2022, Virtual Format.
- Ricketson, L., **Ghosh, D.**, Dorr, M., Dorf M., Hittinger, J., *Continuum Kinetic Simulations of Tokamak Edge Plasmas with COGENT*, Platform for Advanced Scientific Computing (PASC) Conference, Minisymposium on Towards Exascale Computing in Kinetic Simulations of Magnetic Fusion Plasmas. Part III - Advanced Numerical Methods and Algorithms, July 5 - 9, 2021, Virtual Format.
- Ricketson, L., **Ghosh, D.**, Dorr, M., Dorf M., *Block Preconditioning of a Semi-Implicit Gyrokinetic Model of Fusion Plasmas*, 9th International Conference on Computational Methods for Coupled Problems in Science and Engineering (COUPLED PROBLEMS 2021), Minisymposium on Block Preconditioning for Challenging Multiphysics Systems, June 14 - 16, 2021, Virtual Format.
- Ghosh, D.**, Chapman, T., Berger, R. L., Banks, J. W., Copeland, D., *A Semi-Implicit Algorithm for the Simulation of High-Z Plasma Interpenetration*, 15th United States National Congress on Computational Mechanics (USNCCM15), Minisymposium on Innovative Higher Order Discretization Methods in Computational Science and Engineering, in honor of Prof. Mary Wheeler, July 28 - August 1, 2019, Austin, TX.
- Hittinger, J., Dorr, M. R., Dorf, M. A., **Ghosh, D.**, Ricketson, L. F., *Kinetic Edge Plasma Simulation using a Continuum Model*, Platform for Advanced Scientific Computing (PASC), Minisymposium on HPC Challenges in Kinetic Simulations of Plasmas, Part I of III: Eulerian Approach, June 12 - 14, 2019, Zurich, Switzerland.
- Ghosh, D.**, Chapman, T. D., Berger, R. L., Dimits, A., Banks, J. W., *A Multi-Species, Multi-Fluid Model for Simulating Plasma Interpenetration*, SIAM Conference on Computational Science and Engineering, Minisymposium on State-of-the-art high-order Numerical Methods and Complex Fluid Simulations, February 25 - March 1, 2019, Spokane, WA.
- Banks, J. W., Berger, R. L., Brunner, S., Chapman, T. D., Dimits, A., **Ghosh, D.**, *Simulation of Interpenetrating Plasmas in 2D using the Grid Based Continuum Code LOKI*, SIAM Conference on Computational Science and Engineering, Minisymposium on Computational Methods for Kinetic Models of Plasma, February 25 - March 1, 2019, Spokane, WA.
- Ghosh, D.**, Kavouklis, C., Chapman, T., Berger, R. L., Vogman, G. V., *Numerical Simulation of Counterstreaming Plasma Interactions using a Multifluid Model*, 13th World Congress on Computational Mechanics (WCCMXIII), Minisymposium on High-order Discretizations for Multi-physics Applications, July 22 - 27, 2018, New York, NY.
- Dimits, A. M., Berger, R. L., Chapman, T., **Ghosh, D.**, Arrighi W.J., Hittinger, J. A. F., Joseph, I., Kavouklis, C., Brunner, S., Banks, J. W., *Linearized Coulomb Collision Operator for Simulation of Interpenetrating Plasma Streams*, 45th IEEE International Conference on Plasma Science, June 24 - 28, 2018, Denver, CO.
- Implicit-Explicit Time Integration for Multiscale Physics*, Department of Applied Mathematics and Statistics, University of California Santa Cruz, December 2017, Santa Cruz, CA.
- Ghosh, D.**, Constantinescu, E. M., *Characteristic-Based Slow-Wave-Fast-Wave Partitioning for Semi-Implicit Time Integration of Atmospheric Flows*, IMAGE 2017 Theme of the Year: Workshop on Multiscale Geoscience Numerics, National Center for Atmospheric Research (NCAR), May 16 - 19, 2017, Boulder, CO.
- Ghosh, D.**, Dorr, M., Dorf, M., Hittinger, J., *Implementation of Implicit-Explicit Time Integration for the Kinetic Modeling of Tokamak Plasma Edge*, SIAM Conference on Computational Science and Engineering, Minisymposium on Computational Plasma Physics, February 27 - March 3, 2017, Atlanta, GA.
- Dorr, M., Colella, P., Dorf, M., **Ghosh, D.**, Hittinger, J., *Exact Preservation of Zero Velocity Divergence in a High-Order, Mapped-Grid, Finite-Volume Discretization of a Gyrokinetic System*, SIAM Conference on Computational Science and Engineering, Minisymposium on Computational Plasma Physics, February 27 - March 3, 2017, Atlanta, GA.
- Ghosh, D.**, Constantinescu, E. M., *Characteristic-Based Flux Splitting for Implicit-Explicit Time Integration of Low-Mach Number Flows*, 13th U.S. National Congress on Computational Mechanics (USNCCM13), Minisymposium on Advances in Implicit / Explicit (IMEX) Time integration of Multiphysics Systems, July 26 - 30, 2015, San Diego, CA.
- Ghosh, D.**, Constantinescu, E. M., *A Finite-Difference Algorithm with Characteristic-Based Semi-Implicit Time-Integration for the Euler Equations with Gravitational Forcing*, SIAM Conference on Mathematical & Computational Issues in the Geosciences, Minisymposium on Modeling and Simulation of Multiscale and Coupled Processes in Atmospheric Physics, June 29 - July 2, 2015, Stanford, CA.
- Ghosh, D.**, Constantinescu, E. M., *A Compact-Reconstruction WENO Scheme with Semi-Implicit Time Integration*, SIAM Conference on



Computational Science and Engineering, Minisymposium on Recent Advances in High Order Spatial Discretization Methods for PDEs, March 14 - 18, 2015, Salt Lake City, UT.

Ketcheson, D., **Ghosh, D.**, *Stability-Optimized Time Integrators for WENO Discretizations*, SIAM Conference on Computational Science and Engineering, Minisymposium on Advances in Time-stepping Methods, March 14 - 18, 2015, Salt Lake City, UT.

Barajas-Solano, D. A., Tarakovsky, A., **Ghosh, D.**, Constantinescu, E. M., Abhyankar, S., *Probability Density Methods for the Analysis of Power Grids Under Uncertainty*, SIAM Conference on Computational Science and Engineering, Minisymposium on Distributed Cyber-Physical Systems: Modelling and Controlling the Power Grid, March 14 - 18, 2015, Salt Lake City, UT.

**Ghosh, D.**, Constantinescu, E. M., Brown, J., *Scalable Non-Linear Compact Schemes*, International Conference on Spectral and High Order Methods (ICOSAHOM), Minisymposium on Aspects of Time Stepping, June 23 - 27, 2014, Salt Lake City, UT.

*Compact-Reconstruction WENO Schemes - Theory, Implementation and Applications*, National Institute of Aerospace, June 2014, Hampton, VA.

## OTHER CONFERENCES, MEETINGS, TALKS AND POSTERS (55)

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Angus, J. R., Grote, D., **Ghosh, D.**, van de Wetering, J. J., Kemp, A., Vay, J.-L., *Energy-Conserving Implicit Particle-in-Cell Method in WarpX for Dense, Fusion Plasmas*, 2026 IEEE International Conference on Plasma Science (ICOPS), 93.

Lehe, R., Angus, J. R., **Ghosh, D.**, Groenewald, R., Huebl, A., Formenti, A., Grote, D., Myers, A., Shalpoval, O., Vay, J.-L., Zhang, W., Zoni, E., *WarpX for Fusion Research: Recent Advances and New Capabilities*, 67th Annual Meeting of the American Physical Society's Division of Plasma Physics (DPP), November 17 - 21, 2025, Long Beach, CA.

Oudin, A., **Ghosh, D.**, Kur, E., Lancia, L., Riconda, C., Ou, K., Perez-Ramirez, V. M., Cao, S., Singh, D., Redshaw, C., Rajesh, H., Dedeler, P., Edwards, M. R., Michel, P., *Nonlinear Modeling of Photochemically-Induced Gaseous Optical elements*, 13ème Forum Lasers et Plasmas, June 2 - 6, 2025, Le Plan-de-la-Tour, France. Also presented at: 53rd Annual Anomalous Absorption Conference, May 11 - 16, 2025, Sedona, AZ.

Dorf, M., Dorr, M., **Ghosh, D.**, Umansky, M., Geyko, V., Angus, J., Soukhanovskii, V., *Implicit simulations of collisional transport in magnetized plasmas with the continuum gyrokinetic code COGENT*, Sherwood Fusion Theory Conference, April 7 - 9, 2025, New York, NY.

Angus, J. R., van de Wetering, J. J., Farmer, W., Friedman, A., Fu, Y., Geyko, V., **Ghosh, D.**, Grote, D., Larson, D., Zimmerman, G., *An implicit particle code for fully kinetic simulations of burning plasmas*, IFE-STAR Conference, April 7, 2025, Breckenridge, CO.

Terasaki, G. M., **Ghosh, D.**, Navid, A., *Examining Complex Disease Spread Dynamics in Schools and Workplaces Using An Exascale Agent-Based Framework*, SIAM Conference on Computational Science and Engineering, March 3 - 7, 2025, Fort Worth, TX.

Salian, L. V., Rathan, S., **Ghosh, D.**, *A new central compact finite difference scheme with high spectral resolution for dispersive equations*, 19th International Conference on Hyperbolic Problems: Theory, Numerics and Applications, July 1 - 5, 2024, Shanghai, China.

Ricketson, L., Dorr, M., Dorf, M., **Ghosh, D.**, *Solvers and preconditioners enabling flexible IMEX integration of fusion plasmas in COGENT*, 18th Copper Mountain Conference on Iterative Methods, April 14 - 19, 2024, Copper Mountain, CO.

Angus, J., Farmer, W., Friedman, A., Geyko, V., **Ghosh, D.**, Grote, D., Larson, D., Link, A., *An implicit particle code with exact energy and charge conservation for dynamically compressing plasmas: Solver scaling and performance*, 18th Copper Mountain Conference on Iterative Methods, April 14 - 19, 2024, Copper Mountain, CO.

Dorf, M., Dorr, M., **Ghosh, D.**, Angus, J. R., *Modeling tokamak edge plasma with the continuum full-F code COGENT*, 65th Annual Meeting of the American Physical Society's Division of Plasma Physics (DPP), October 30 - November 3, 2023, Denver, CO.

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